

Technical Report: Telco Customer Churn Prediction Using Logistic Regression

1. Executive Summary

This report details the development and evaluation of a predictive model designed to identify customers at risk of churning from a telecommunications service.

2. Dataset Overview & Preprocessing

2.1 Data Characteristics

- **Total Observations:** 7,043 rows
- **Total Initial Features:** 21 columns (18 categorical, 3 numerical, and 1 unique identifier)
- **Target Variable:** Churn (Yes / No)

2.2 Data Cleaning and Feature Engineering

- **Data Type Correction:** The TotalCharges column contained blank spaces masquerading as non-null values (causing it to load as an object type). These values—representing brand-new customers with 0 tenure—were coerced into numeric form, and missing values were filled with 0.
- **Feature Removal:** customerID was dropped as it possesses no predictive power.
- **Categorical Encoding: * Binary Mapping:** Binary variables (Partner, Dependents, PhoneService, PaperlessBilling, Churn) were mapped directly to 1 (Yes) and 0 (No).
 - **Ordinal/Service Alignment:** Internet service-related features (OnlineSecurity, OnlineBackup, etc.) and MultipleLines were unified to a 0/1 schema, mapping text variations like "No internet service" or "No phone service" to 0.
 - **One-Hot Encoding:** Multi-category nominal columns (gender, InternetService, Contract, PaymentMethod) were converted into dummy variables using drop_first=True to prevent multicollinearity.

2.3 Data Partitioning & Scaling

The dataset was split into an **80/20** train-test split to ensure rigorous evaluation:

3. Model Performance and Evaluation

A baseline **Logistic Regression** model was trained on the standardized data. The model's predictions on the unseen test set yielded the following metrics:

3.1 Global Metrics

- **Classification Accuracy:** 82.11%
- **ROC AUC Score:** 0.86

3.2 Confusion Matrix Breakdown

	Predicted Non-Churn (0)	Predicted Churn (1)
Actual Non-Churn (0)	933 (True Negative)	103 (False Positive)
Actual Churn (1)	149 (False Negative)	224 (True Positive)

3.3 Detailed Classification Report

Plaintext

	precision	recall	f1-score	support
0	0.86	0.90	0.88	1036
1	0.69	0.60	0.64	373
accuracy				0.82 1409
macro avg				0.77 0.75 0.76 1409
weighted avg				0.82 0.82 0.82 1409

- **Non-Churn Class (0):** Strong recall (90%) and precision (86%), showing the model is highly effective at identifying stable customers.
- **Churn Class (1):** Moderate recall (60%) and precision (69%). The F1-score of 0.64 highlights a minor trade-off where the model misses roughly 40% of actual churning customers (False Negatives).

4. Feature Importance & Operational Insights

4.1 Top Drivers of Customer Churn (Positive Coefficients)

Features that heavily increase the likelihood of a customer leaving:

1. **TotalCharges = +0.661**
2. **InternetService_Fiber optic = +0.533**
3. **StreamingMovies = +0.195**

4.2 Top Drivers of Retention (Negative Coefficients)

Features that protect against customer churn:

1. **Tenure = -1.360**
2. **Contract_Two year = -0.611**
3. **InternetService_No = -0.455**
4. **MonthlyCharges = -0.418**